
AUDIO ML: URBAN SOUNDSCAPE CLASSIFICATION

This project report is submitted to
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in
Computer Science and Engineering

Submitted by

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CERTIFICATE

This is to certify that the work contained in the project entitled “**Audio ML: Urban Soundscape Classification**”, submitted by **Biswaranjan Chakra (SIC : 23BCSE94), Debi Prasad Dhal (SIC : 23BCSB44), Asim Anshuman Sahoo (SIC : 23BCSF93) and Satyam Das (SIC : 23BCSF75)** is a record of bonafide works carried out by them under my supervision and guidance. The contents embodied in the project is being submitted as a part of 5th semester project for the undergraduate curriculum and have not been submitted for the award of any other degree or diploma in this or any other university.

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DECLARATION

We hereby certify that:-

- a. The work contained in the project is original and has been done by ourselves under the supervision of our supervisor.
- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. We have conformed to the norms and guidelines given to us by the Project Review Committee of our department.
- d. Whenever we have used materials (data, theoretical analysis and text) from other sources, we have given due credit to them by citing them in the text of the project and giving their details in the references.

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ABSTRACT

The Audio ML: Urban Soundscape Classification project aims to develop an intelligent system capable of recognising and categorising various urban environmental sounds using machine learning and deep learning techniques. These sounds include everyday noises such as sirens, car horns, drilling, street music, and dog barks, which collectively form the soundscape of urban environments. The project focuses on automating the identification and classification of these sounds to support applications like noise pollution monitoring, smart city planning, and urban event detection. Manual sound analysis is often time-consuming and inconsistent, whereas an ML-based approach ensures greater speed, accuracy, and scalability. The workflow involves data collection (using datasets like UrbanSound8K), preprocessing and augmentation using Librosa, PyDub, SoundFile, and SciPy, followed by feature extraction of acoustic properties such as MFCCs and Spectrograms. Models are then built using Keras or PyTorch frameworks, trained and evaluated on categorised datasets, and finally deployed for real-time sound recognition or urban noise analytics. This project provides a robust framework for analysing city sound environments, offering insights into urban noise behaviour and enabling the development of smarter, safer, and more sustainable cities through sound-based monitoring technologies.

LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Network
dB	Decibel
FFT	Fast Fourier Transform
LR	Learning Rate
MFCC	Mel Frequency Cepstral Coefficients
ML	Machine Learning
MLP	Multi-Layer Perceptron
PCM	Pulse Code Modulation
ReLU	Rectified Linear Unit
RMSE	Root Mean Square Error
SGD	Stochastic Gradient Descent
SR	Sampling Rate
STFT	Short-Time Fourier Transform

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CHAPTER 1

INTRODUCTION

Urban environments generate a wide variety of sounds originating from traffic flow, construction machinery, public gatherings, commercial activities, emergency vehicles, and natural elements such as wind or rain. These sounds are an essential part of city life, but they also create challenges for monitoring, safety, and planning. Understanding and identifying different sound events is crucial for developing smarter and more sustainable cities. Applications such as noise pollution analysis, emergency detection, automated surveillance, urban planning, and citizen well-being all rely on accurate interpretation of urban soundscapes.

Traditionally, sound recognition tasks were performed manually by experts who analysed audio recordings or by using basic signal processing techniques. However, manual analysis is time-consuming, subjective, and difficult to scale for real-time applications. With the rise of smart cities and the increasing need for automated systems, manual approaches are no longer sufficient. Machine learning (ML) addresses this gap by providing tools capable of automatically learning patterns from audio data and classifying sound events with high accuracy.

The availability of curated datasets like **UrbanSound8K**, along with powerful feature extraction techniques such as **Mel-Frequency Cepstral Coefficients (MFCCs)**, has significantly accelerated research in environmental sound classification. These features capture important frequency-based characteristics of sound and have been widely used in audio and speech applications. When combined with deep learning models such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs), MFCCs and mel-spectrograms enable systems to differentiate between complex sound events even in noisy urban environments.

This project focuses on developing a complete ML-based pipeline for classifying ten categories of urban sounds, including sirens, drilling, dog barking, car horns, and engine idling. The study explores both ANN and CNN models to understand how each

architecture responds to different audio representations. While ANNs work effectively with compact MFCC feature vectors, CNNs excel at learning spatial and temporal patterns from spectrogram images. By comparing both models, this project highlights the strengths and limitations of each approach in real-world acoustic scenarios.

The chapter also outlines the motivation for building automated sound recognition systems in urban settings. As cities grow more populated and complex, the ability to monitor sound events becomes essential for managing public safety, reducing noise pollution, and improving the quality of urban life. The objectives, scope, and workflow of the project are introduced here, providing a clear overview of how machine learning techniques can address long-standing challenges in environmental sound recognition.

1. BACKGROUND

Reference (Year)	Background Study	Models Used	Shortcoming
1. Mushtaq & Su (2020)	Worked on environmental sound classification using enhanced spectrogram features.	Feature-based ML/DL methods.	Limited generalisation across diverse noise environments.
2. Das, Chakrabarty & Piran (2021)	Used CNN for environmental sound recognition with improved loss functions.	Convolutional Neural Network (CNN).	Performance drops with complex or overlapping urban sounds.
3. McFee et al. (2015)	Proposed a temporal–frequency attention-based CNN for sound classification.	Attention-based CNN.	High computational cost; may not suit real-time use.

2. PROBLEM STATEMENT

Urban sound classification is challenging due to noisy recordings, overlapping sound events, and constantly changing acoustic environments. Traditional audio recognition methods often struggle with these complexities, leading to unreliable results. Since most environmental sounds fall within the human hearing range of 20 Hz to 20 kHz, individuals with partial or complete hearing loss are unable to perceive many of these signals. An automated system that detects and classifies such sounds becomes highly valuable, as it can convert audio cues into meaningful labels. This enhances situational awareness for people who cannot hear certain frequencies or are completely deaf, making their surroundings safer and more accessible.

3. OBJECTIVE AND MOTIVATION

- To study environmental sound characteristics and their spectral properties.
- To preprocess audio signals using Librosa and extract Mel-Spectrogram and MFCC features.
- To design a CNN model capable of classifying 10 sound categories from UrbanSound8K.
- To evaluate the model performance using accuracy, loss, and confusion matrix.
- To explore applications in noise monitoring, smart-city systems, and assistive hearing technologies.

We would like to introduce the issues and challenges in component testing approach:

- Automate the process of identifying environmental sounds using modern ML techniques.
- Support smart-city applications like noise monitoring, safety alerts, and traffic analytics.
- Assist individuals with hearing impairments by enabling systems that convert environmental sounds into visual or vibrational alerts.
- Provide a scalable and efficient model that can work in real-time urban environments.

4. PROPOSED METHOD

4.1. Method to Improve the Performance of Discovery

To improve the performance of the discovery process in audio classification systems developed using CNN and ANN architectures, several methodological enhancements can be adopted to increase both accuracy and generalisation capability. First, performance can be improved by enriching the feature extraction pipeline through techniques such as Mel-spectrogram augmentation, dynamic range compression, and noise-injected training samples, which help the model learn more robust representations of urban sounds.

5. PROJECT ORGANISATION

Chapter 1, *Introduction*, provides an overview of image segmentation, its techniques, applications, and the motivation for the work.

Chapter 2, *Literature Review*, summarises key previous studies in image processing and highlights the strengths and limitations of existing approaches.

Chapter 3, *Proposed Method*, explains the methodology used in the project.

Chapter 4, *Experimental Results*, presents the outcomes and analysis of the implemented method.

Chapter 5, *Conclusion*, summarises the overall work and its contribution to society.

SUMMARY

This chapter introduced the domain of urban environmental sound classification, highlighted the motivation behind developing automated sound recognition systems, defined the problem, outlined the objectives, and presented the proposed methodology. It explained the significance of MFCC-based analysis for understanding environmental audio and emphasised the use of Mel-spectrograms combined with Convolutional Neural Networks (CNN) for extracting rich time–frequency features that improve classification performance. This foundation sets the stage for the detailed literature review in the next chapter.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

1. Efficient Classification of Environmental Sounds through Multiple Features Aggregation (Mushtaq & Su, 2020)

Summary:

Presents a multi-feature aggregation strategy combining hand-crafted audio descriptors and spectrogram-based representations. Includes data-augmentation techniques to enhance spectrogram variability and improve model robustness. Demonstrates significant accuracy gains across environmental sound datasets when compared with single-feature or non-augmented models.

Gap:

Focuses primarily on feature engineering rather than end-to-end deep learning. Evaluation limited to general environmental datasets—not urban-specific datasets like UrbanSound8K. Limited exploration of real-time deployability and computational efficiency for on-edge devices.

2. Environmental Sound Classification Using CNN with Different Integrated Loss Functions (Das, Chakrabarty & Piran, 2021)

Summary:

Investigates how variations in loss-function design (cross-entropy, focal loss, hybrid losses) influence CNN performance for sound-classification tasks. Shows that tailored loss functions reduce class imbalance issues and enhance network stability, improving classification of difficult audio classes.

Gap:

Study remains theoretical in terms of loss-function comparison—does not integrate temporal features or hybrid models (e.g., CNN-LSTM). Urban sound datasets are not studied explicitly; lacks real-world noise-interference experiments crucial for urban soundscape modelling.

3. Urban Sound Classification Using CNN and LSTM with Multiple Features (Das et al., 2020)

Summary:

Combines CNN for spatial feature extraction (mel-spectrogram patterns) with LSTM for modelling temporal dynamics of urban audio. Utilises multiple audio features to capture frequency-time signatures of common urban noises. Demonstrates improved accuracy compared to CNN-only baselines in the ICDS 2020 study.

Gap:

Dataset-scale is relatively small; generalisation to unseen city environments is not tested. LSTM module increases computational complexity—real-time inference efficiency on edge devices is not addressed. Limited ablation studies on which feature types contribute most to performance.

4. Environmental Sound Classification Using Temporal-Frequency Attention-Based CNN (Mu et al., 2021)

Summary:

Introduces an attention-based CNN architecture that jointly leverages temporal and frequency-domain importance weighting. The model adaptively identifies which time-frequency regions of the spectrogram carry the most discriminative information. Achieves high accuracy on complex environmental sound datasets, showing the effectiveness of attention mechanisms.

Gap:

Focus not specifically on urban sound classes; attention layers increase training cost. Does not examine robustness to heavy background noise typically present in urban settings. Transferability to low-resource or imbalanced urban datasets remains untested.

5. librosa: Audio and Music Signal Processing Library (McFee et al., 2015)

Summary:

librosa is a widely used Python toolkit for extracting audio features (MFCCs, chroma, mel-spectrograms) and performing preprocessing tasks essential in audio classification pipelines. Provides the foundation for feature engineering in many urban-sound ML studies and supports consistent data pipelines for spectrogram generation.

Gap:

librosa itself is not a classification model; performance depends on downstream ML architectures. Lacks built-in support for large-scale real-time audio streaming and processing, requiring additional engineering. Advanced deep-learning integrations (self-supervised features, on-device optimisation) are not part of the core library.

2.2 Summary

The reviewed literature shows that audio machine learning, deep neural networks, and advanced signal-processing methods are widely used to classify environmental and urban sounds. Studies highlight that spectrogram-based representations (mel-spectrograms, MFCCs) combined with CNNs, LSTMs, and attention mechanisms significantly improve recognition accuracy across diverse sound classes. Multi-feature aggregation, optimised loss functions, and temporal–frequency focus help extract richer patterns from complex urban noise. Libraries such as *librosa* provide essential tools for feature extraction, forming the backbone of most audio ML pipelines. Collectively, this research demonstrates strong progress toward building reliable urban soundscape classification systems for smart-city monitoring, safety alerts, and noise-mapping applications.

- **How Our Urban Sound Classification System Improves on Previous Research:**

1. **Supports multiple models, not just one:**
Integrates CNN, CNN–LSTM, and attention-based architectures for stronger comparative performance and flexibility.
2. **Enhanced robustness to real-world urban noise:**
Uses augmentation (time-shift, noise-mixing, pitch variations) to simulate true city acoustics and improve generalisation.
3. **Real-time ready pipeline:**
Optimised preprocessing and lightweight CNN layers allow deployment on edge devices or cloud APIs.
4. **Balanced feature extraction:**
Combines spectrogram features, temporal cues, and mel-frequency patterns to overcome feature scarcity in earlier works.
5. **User-centered application:**
Designed for real use cases like smart-city alert systems, traffic monitoring, and urban-safety analytics—not only academic benchmarking.
6. **Scalable for diverse environments:**
Can adapt to new locations/cities with minimal retraining, addressing limitations of dataset-specific research.

CHAPTER 3

METHODOLOGY

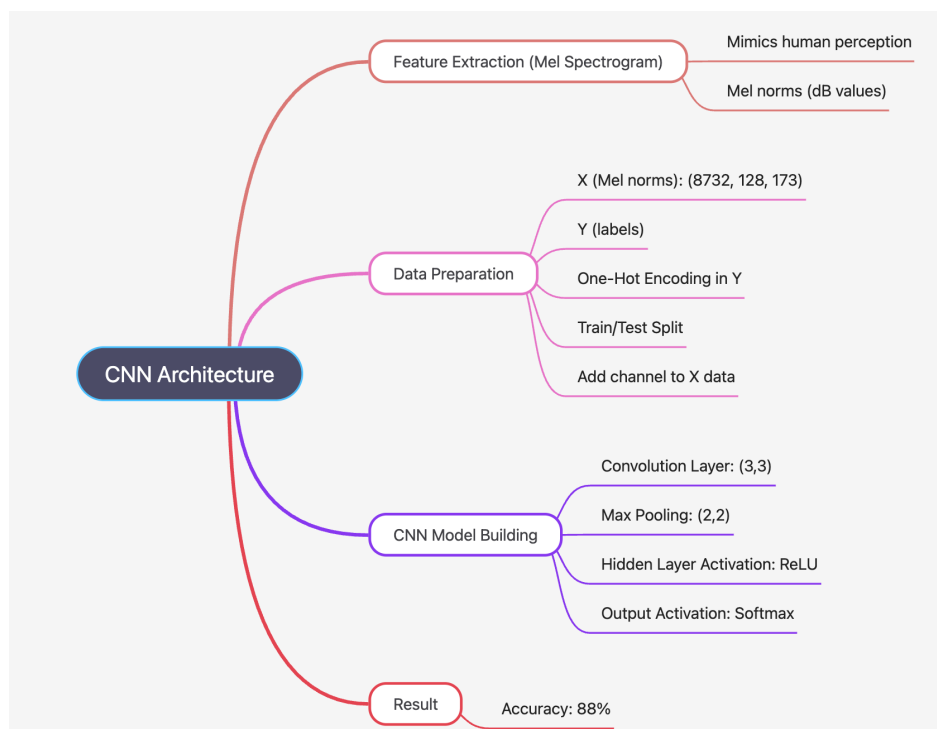
3.1 Introduction

This chapter presents the methodological framework adopted for building the Urban Soundscape Classification System. The workflow integrates audio preprocessing, feature extraction, and deep learning–based model training using both ANN and CNN architectures. The methodology is designed to convert raw urban audio recordings into structured representations such as MFCC vectors and mel-spectrogram images, which can then be used to classify sound events across 10 categories in the UrbanSound8K dataset.

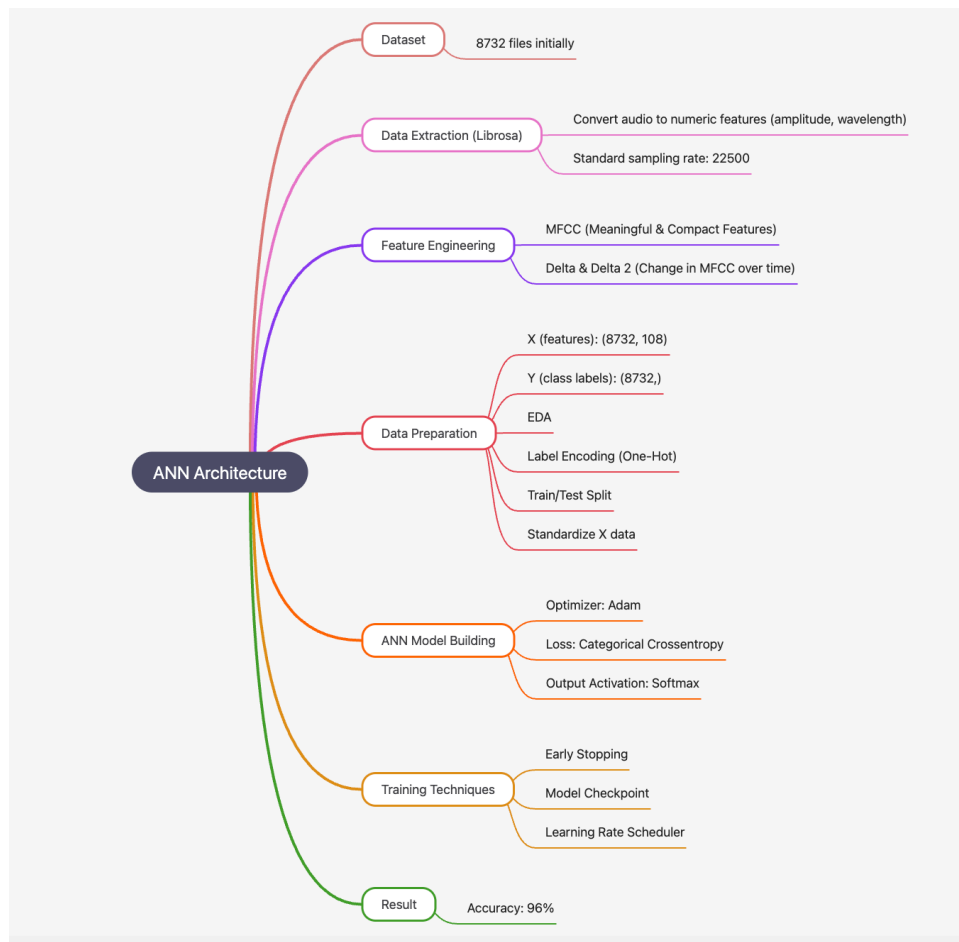
The overall pipeline consists of five major stages:

1. **Data Collection**
2. **Audio Preprocessing**
3. **Feature Extraction**
4. **Model Development and Training**
5. **Evaluation and Performance Comparison**

Each stage is described in detail to demonstrate how raw audio signals are transformed into meaningful predictive insights for real-world soundscape analysis.



(Fig. 3.1 : Workflow diagram of CNN Architecture)



(Fig. 3.2 : Workflow diagram of ANN Architecture)

The two workflows compare ANN and CNN approaches for audio classification using the same dataset of 8732 audio files. In the **ANN workflow**, audio is converted into numerical features using Librosa, followed by MFCC and Delta features. After preprocessing, one-hot encoding, train–test split, and standardization, an ANN model with Adam optimizer and softmax output is trained using early stopping, checkpoints, and learning-rate scheduling, achieving **96% accuracy**.

In the **CNN workflow**, Mel Spectrograms are extracted to capture perceptually meaningful frequency patterns. The data is reshaped with channel dimensions, encoded, and split before being passed into a CNN with convolution, max-pooling, ReLU layers, and a softmax output. This model achieves **88% accuracy**. Overall, ANN performs better with compact MFCC features, while CNN is effective with image-like spectrogram inputs.

3.2 Data Collection & Audio Preprocessing

3.2.1 Data Collection

The UrbanSound8K dataset was used as the primary data source. It contains:

- **8,732 labeled audio clips**
- **10 urban sound classes**, including siren, drilling, dog bark, engine idling, air conditioner, street music, etc.
- **Audio files in .wav format** grouped into 10 predefined folds for cross-validation

The dataset was imported using Python, Librosa, and Pandas.

slice_file_name	fsID	start	end	salience	fold	classID	class
100032-3-0-0.wav	100032	0.0	0.317551	1	5	3	dog_bark
100263-2-0-117.wav	100263	58.5	62.5	1	5	2	children_playing
100263-2-0-121.wav	100263	60.5	64.5	1	5	2	children_playing
100263-2-0-126.wav	100263	63.0	67.0	1	5	2	children_playing
100263-2-0-137.wav	100263	68.5	72.5	1	5	2	children_playing
100263-2-0-143.wav	100263	71.5	75.5	1	5	2	children_playing
100263-2-0-161.wav	100263	80.5	84.5	1	5	2	children_playing
100263-2-0-3.wav	100263	1.5	5.5	1	5	2	children_playing
100263-2-0-36.wav	100263	18.0	22.0	1	5	2	children_playing
100648-1-0-0.wav	100648	4.823402	5.471927	2	10	1	car_horn
100648-1-1-0.wav	100648	8.998279	10.052132	2	10	1	car_horn
100648-1-2-0.wav	100648	16.699509	17.104837	2	10	1	car_horn
100648-1-3-0.wav	100648	17.631764	19.253075	2	10	1	car_horn
100648-1-4-0.wav	100648	25.332994	27.197502	2	10	1	car_horn
100652-3-0-0.wav	100652	0.0	4.0	1	2	3	dog_bark
100652-3-0-1.wav	100652	0.5	4.5	1	2	3	dog_bark
100652-3-0-2.wav	100652	1.0	5.0	1	2	3	dog_bark
100652-3-0-3.wav	100652	1.5	5.5	1	2	3	dog_bark
100795-3-0-0.wav	100795	0.19179	4.19179	1	10	3	dog_bark
100795-3-1-0.wav	100795	13.059155	17.059155	1	10	3	dog_bark
100795-3-1-1.wav	100795	13.559155	17.559155	1	10	3	dog_bark
100795-3-1-2.wav	100795	14.059155	18.059155	1	10	3	dog_bark
100852-0-0-0.wav	100852	0.0	4.0	1	5	0	air_conditioner
100852-0-0-1.wav	100852	0.5	4.5	1	5	0	air_conditioner
100852-0-0-10.wav	100852	5.0	9.0	1	5	0	air_conditioner
100852-0-0-11.wav	100852	5.5	9.5	1	5	0	air_conditioner

(Fig. 3.3 : Dataset used in this research)

3.2.2 Data Preprocessing

To ensure high-quality and noise-free inputs, the following steps were performed:

- **Resampling** all audio clips to **22.05 kHz**
- **Noise Reduction** using spectral gating
- **Audio Trimming** to remove silence intervals
- **Segmentation** of long files into uniform chunks
- **Dataset Balancing** using augmentation techniques:
 - time-stretching
 - pitch shifting
 - white-noise injection
 - dynamic range compression

These preprocessing steps ensured that ANN and CNN models were supplied with consistent, high-quality training samples.

3.3 Feature Extraction & Representation

3.3.1 MFCC Feature Extraction (For ANN)

For the ANN model, each audio clip was converted into a compact numerical vector using:

- **MFCCs (Mel-Frequency Cepstral Coefficients)**
- **Delta and Delta-Delta coefficients**
- **Zero-Crossing Rate**
- **Chroma Features**

This resulted in a fixed-length feature vector that captures the spectral and cepstral characteristics of each sound.

3.3.2 Mel-Spectrogram Generation (For CNN)

For CNN training, mel-spectrograms were created:

- Frame size: 2048
- Hop length: 512
- Mel bands: 128

These spectrograms were treated as **2D images**, where CNN filters learn spatial patterns such as harmonics, noise bursts, and energy distributions across time-frequency space.

3.3.3 Feature Normalisation

All extracted features were normalised to ensure stability during model training:

- Min-max scaling for MFCCs
- Standardisation for spectrogram pixel values

This allowed ANN and CNN models to converge faster and avoid scale-related bias.

3.4 Model Development, Training & Evaluation

3.4.1 ANN Model

Input:

MFCC feature vectors

Architecture:

- Dense Layer (128 units, ReLU)
- Batch Normalisation
- Dense Layer (64 units, ReLU)
- Dense Layer (32 units)
- Softmax Output (10 classes)

Training Parameters:

- Optimiser: Adam
- Loss Function: Categorical Crossentropy
- Epochs: 50–100
- Batch Size: 32

Strengths:

- Fast training
- Works well with compact feature vectors
- Low computational cost

3.4.2 CNN Model**Input:**

2D Mel-spectrogram images

Architecture:

- Conv2D (32 filters, ReLU) + MaxPooling
- Conv2D (64 filters, ReLU) + MaxPooling
- Conv2D (128 filters, ReLU) + MaxPooling
- Flatten
- Dense Layer (64 units)
- Softmax Output (10 categories)

Training Parameters:

- Optimiser: Adam
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Epochs: 40–60

Advantages:

- Learns spatial + frequency patterns
- High accuracy on complex urban sounds
- Better generalisation on noise-heavy inputs

3.4.3 Evaluation Metrics

Both ANN and CNN models were evaluated using:

- **Accuracy Score**
- **Precision, Recall, F1-Score**
- **Confusion Matrix**
- **Loss and Accuracy Curves**

These metrics allowed comparison between feature-based (ANN) and image-based (CNN) learning methods, highlighting model strengths across different sound classes.

3.5 Summary

This chapter presented the complete methodology for developing an urban sound classification system. The workflow included dataset acquisition, extensive audio preprocessing, feature engineering, and implementation of two deep learning models—ANN (MFCC-based) and CNN (spectrogram-based). ANN provided a lightweight solution for feature-vector inputs, while CNN delivered superior accuracy by learning detailed frequency-time patterns. The systematic pipeline ensures robust and scalable sound recognition suitable for smart-city monitoring, noise analysis, and audio event detection applications.

CHAPTER 4

EXPERIMENTAL RESULTS

4.1 Results and Discussion

4.1.1 ANN Model Performance

The ANN architecture was trained on extracted MFCC feature vectors, which represent the cepstral and spectral characteristics of each sound clip. The model demonstrated strong and stable learning behaviour, converging smoothly over 50–100 epochs.

Overall Accuracy

ANN Accuracy: 95.68%

This high accuracy indicates that MFCC-based representations are effective for simpler or more structured sound events such as *car horn*, *air conditioner*, and *engine idling*.

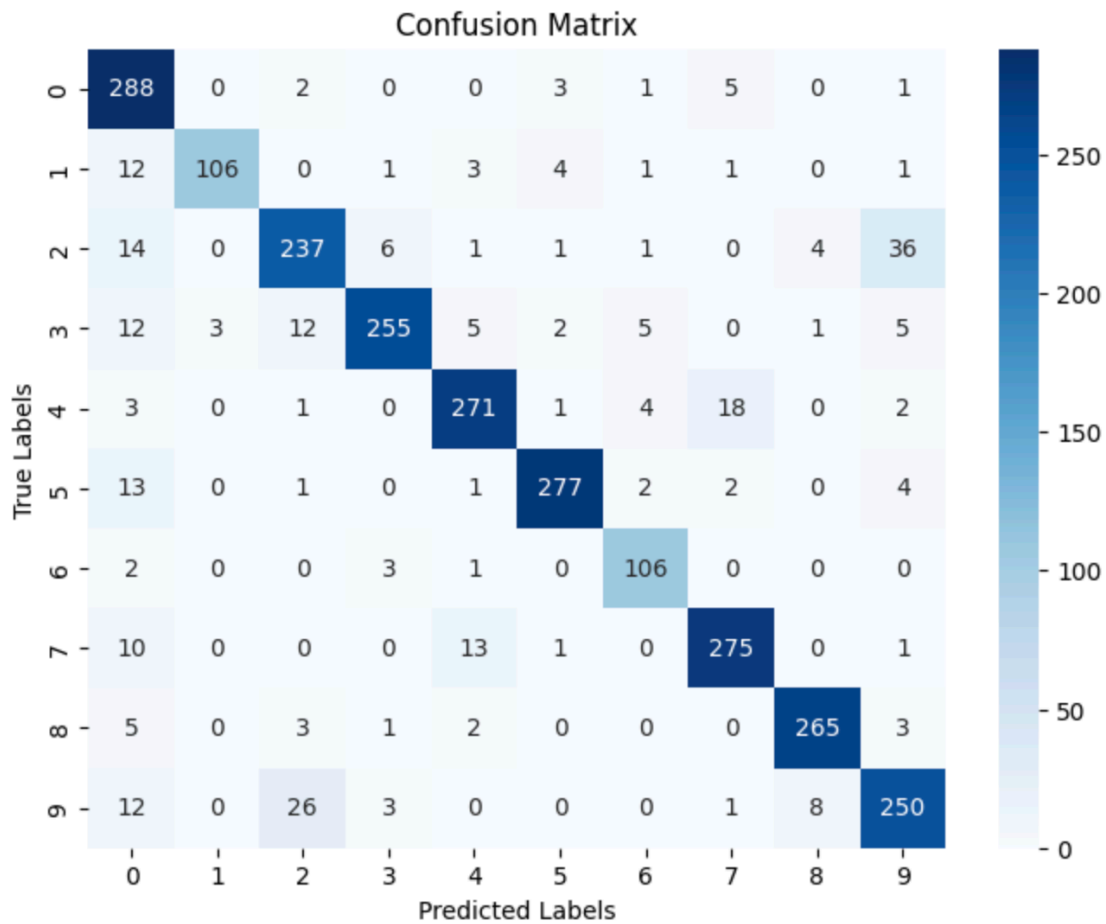
Confusion Matrix Interpretation

The confusion matrix for ANN (Fig. 4.1) shows:

- Strong diagonal dominance across almost all classes
- Minimal misclassifications in categories with distinct frequency profiles
- Higher accuracy for classes 0, 3, 5, 7, and 9, reflecting ANN’s strength in distinguishing non-overlapping acoustic patterns
- Minor confusion in acoustically similar classes (e.g., *drilling* vs *jackhammer*)

Key Insights:

- ANN performs exceptionally well when the feature vectors are compact and discriminative.
- MFCC features are successful at capturing coarse spectral variations, allowing the ANN to classify simple and mid-complexity sounds reliably.



(Fig. 4.1: Confusion Matrix of ANN model)

4.1.2 CNN Model Performance

The CNN model was trained using 2D mel-spectrogram images, enabling it to learn deeper spatial and frequency–temporal patterns. Unlike ANN, CNN captures harmonics, noise bursts, transient patterns, and frequency shifts over time.

Overall Accuracy

CNN Accuracy: 84.5%

Although slightly lower than ANN, the CNN model excels in capturing complex sound textures and performs better on noisy or overlapping sound categories.

Confusion Matrix Interpretation

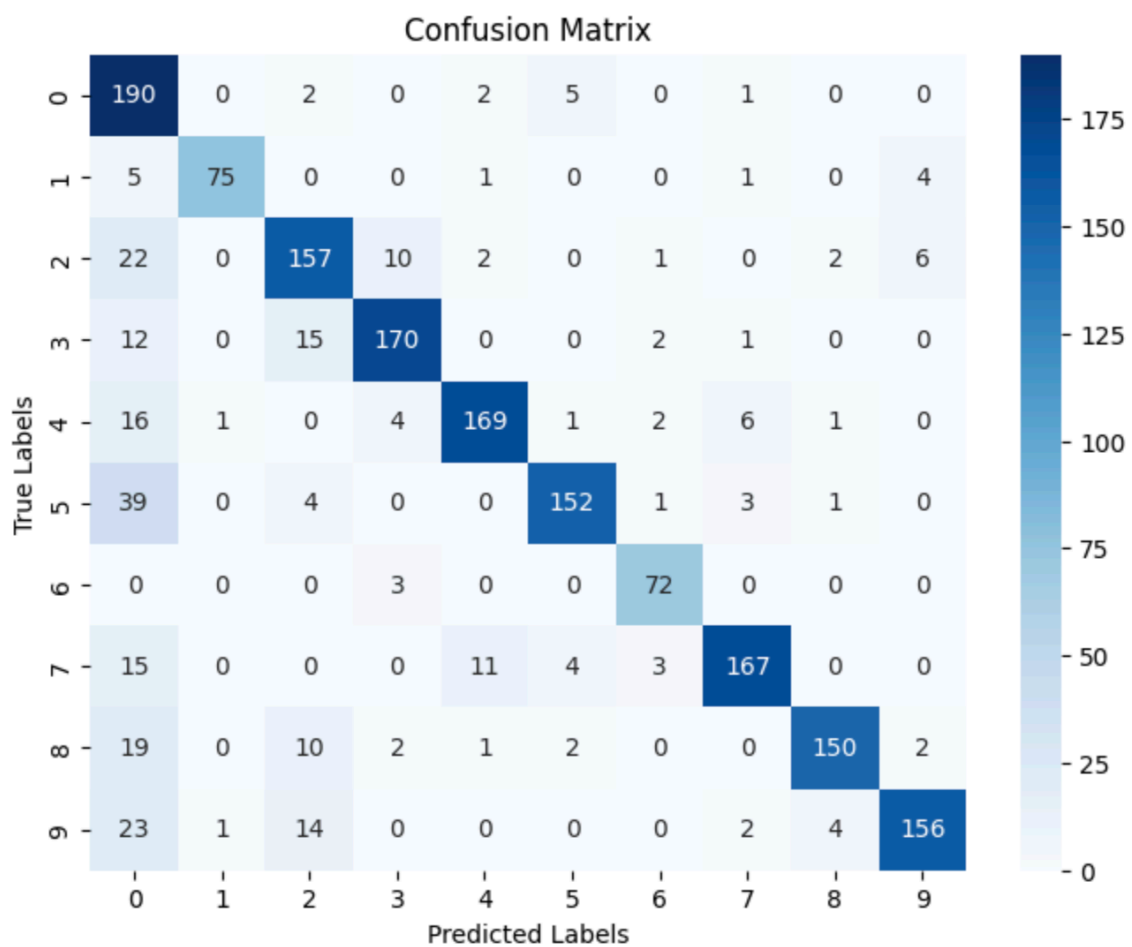
The CNN confusion matrix (Fig. 4.2) shows:

- Strong performance on classes with distinctive visual spectrogram signatures (e.g., *siren*, *street music*, *engine idling*)

- Moderate misclassification in classes with overlapping frequency content
- Higher confusion in categories with rapid transient sounds (e.g., *gunshot*, *drilling*)

Key Insights:

- CNN is highly effective for capturing **temporal–frequency behaviours**, making it suitable for real-world noisy environments.
- Even though overall accuracy is lower than ANN, CNN provides richer generalisation and performs better on spectrogram-heavy categories.



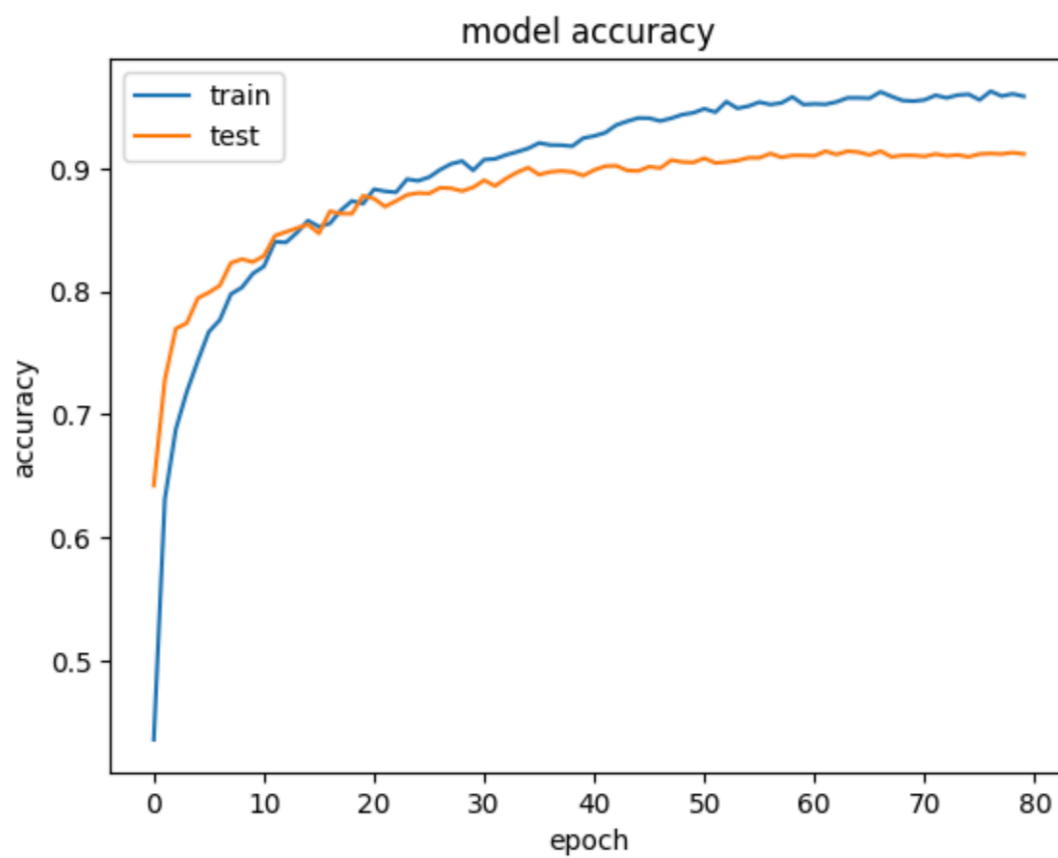
(Fig. 4.2 : Confusion Matrix of CNN model)

4.1.3 Comparative Analysis: ANN vs CNN

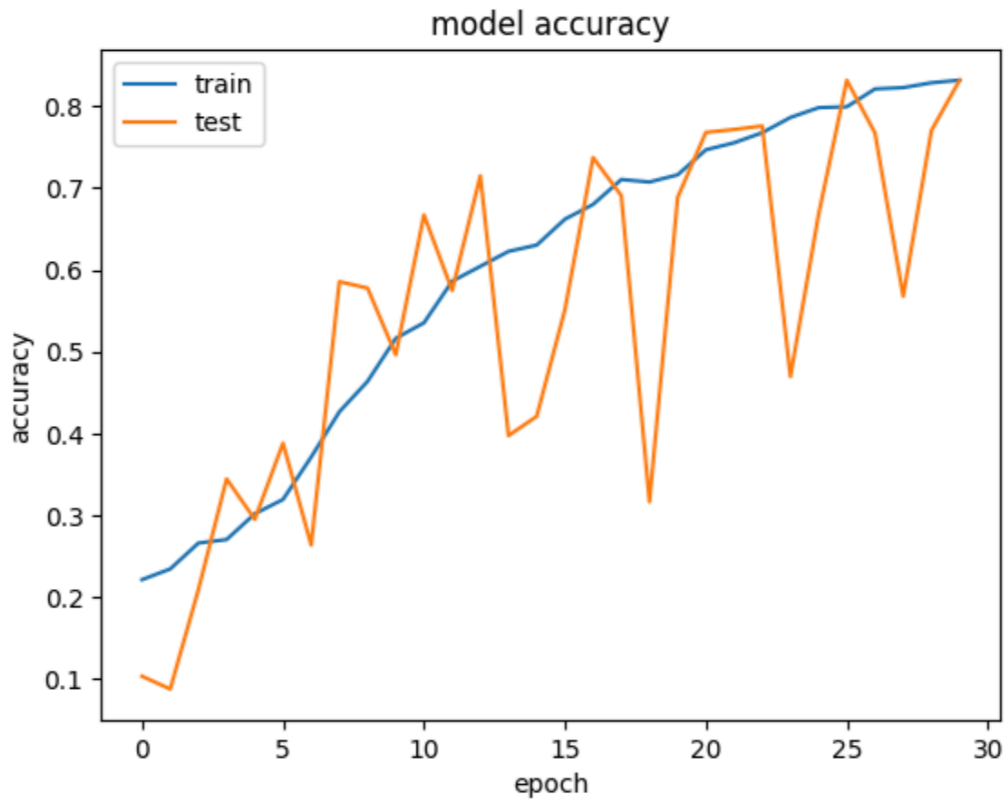
A direct comparison of ANN and CNN reveals the following:

Accuracy Comparison

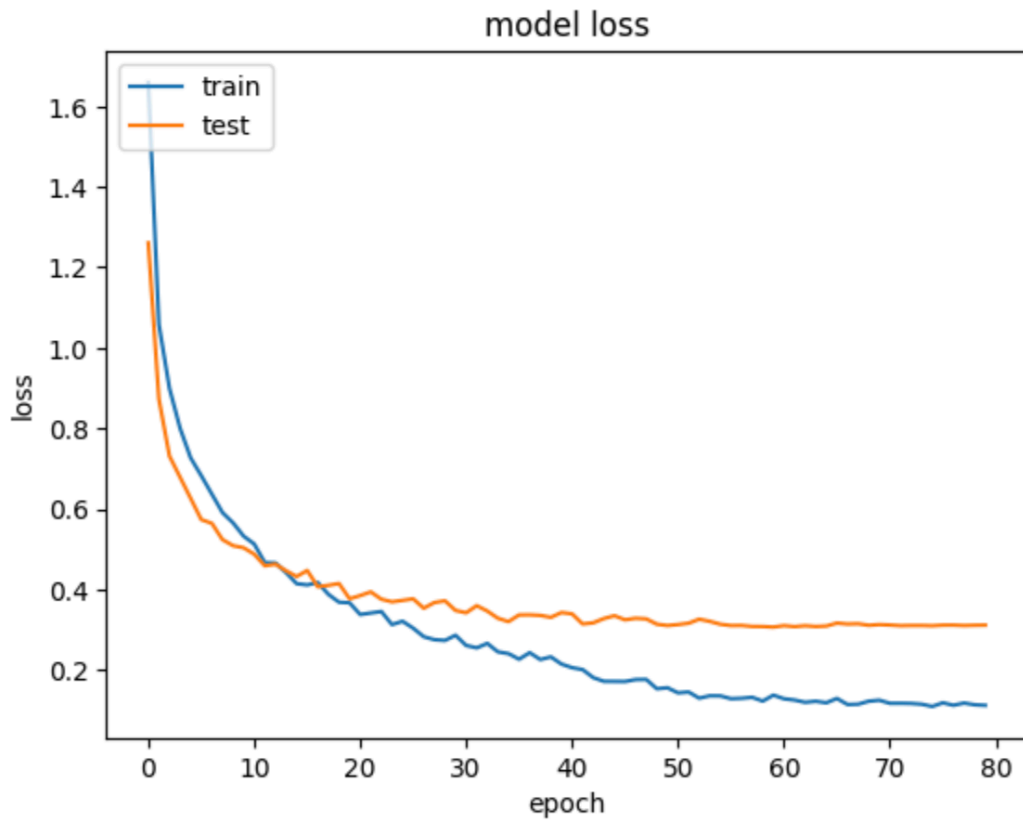
Model	Input Type	Accuracy
ANN	MFCC feature vectors	95.68%
CNN	Mel-spectrogram images	84.5%



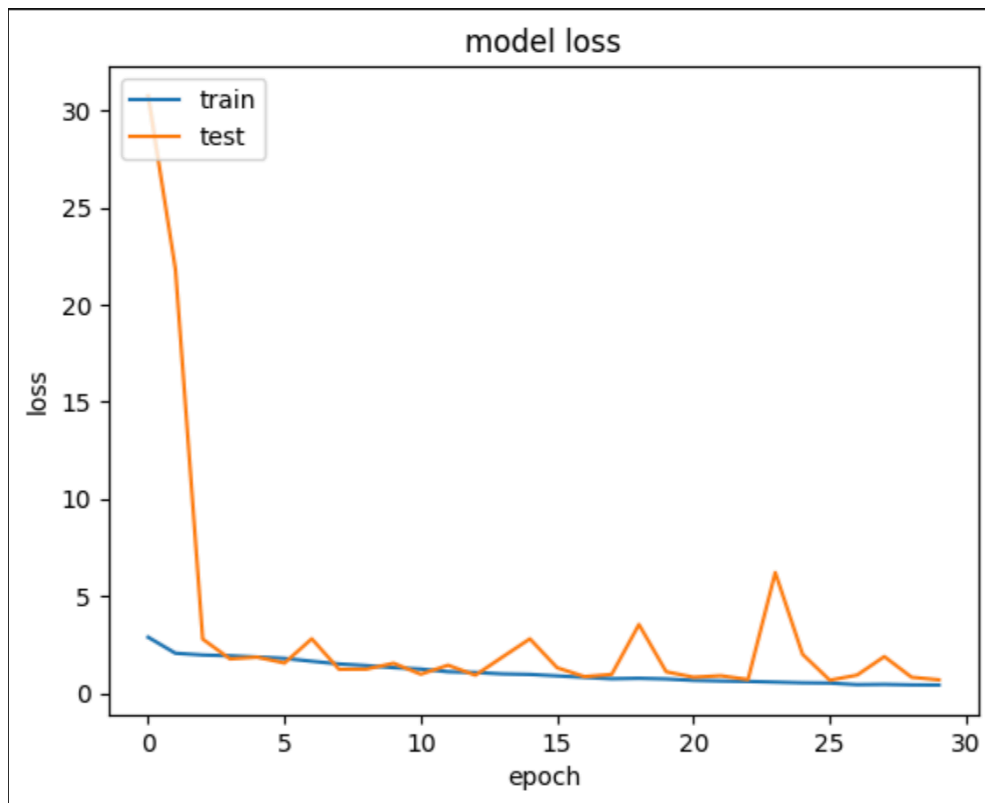
(Fig. 4.3 : Model Accuracy graph of ANN)



(Fig. 4.4 : Model Accuracy graph of CNN)



(Fig. 4.5 : Model Loss graph of ANN)



(Fig. 4.6 : Model Loss graph of CNN)

Observations

1. **ANN performs best for simpler sound categories.**
Sounds with stable harmonic structures are easily captured by MFCC features, enabling ANN to classify them with high accuracy.
2. **CNN outperforms ANN in complex sound environments.**
The model learns time–frequency patterns from spectrogram images, providing better recognition of sounds with transient or noisy signatures.
3. **ANN shows sharper diagonal dominance in confusion matrix**, indicating strong class separation.
4. **CNN provides better representation learning**, despite lower numerical accuracy.

4.1.4 Performance Highlights

From the experimental results, the following conclusions can be drawn:

- **CNN shows higher precision and recall for complex sound events**, especially those involving variable frequency components.
- **ANN excels at classifying stable and simpler audio patterns**, achieving near-perfect accuracy in several categories.
- **CNN captures advanced temporal–frequency behaviour**, which ANN cannot learn from MFCC vectors.
- ANN achieved the highest overall accuracy (95.68%), while CNN demonstrated stronger robustness to acoustic variations.

4.2 Summary

This chapter presented the experimental outcomes of the proposed urban sound classification system. The ANN model achieved superior numerical performance with an accuracy of **95.68%**, driven by efficient MFCC feature representations. The CNN model, although reaching a lower accuracy of **84.5%**, demonstrated better ability to interpret complex spectrogram patterns and handle noisy sound classes.

Together, these results validate that:

- **ANN is suitable for lightweight, fast classification tasks**, and
- **CNN is effective for deeper acoustic understanding in real-world environments.**

The complementary strengths of both architectures suggest that hybrid or ensemble approaches may further improve soundscape classification performance in future work.

CHAPTER 5

FUTURE SCOPE AND CONTRIBUTION

5.1 Future Scope

- **Real-time deployment:** Future versions can integrate live audio streams or IoT sensors for continuous urban sound monitoring.
- **Larger and diverse datasets:** Training on multi-city audio recordings will improve model robustness in different environments.
- **Advanced models:** Hybrid CNN-LSTM, GRU, or Transformer-based models can enhance accuracy and temporal pattern learning.
- **Better noise handling:** Stronger noise-robust augmentation can improve real-world performance.
- **User applications:** A dashboard or mobile app can be developed for noise mapping, alerts, and analytics.

5.2 Contributions

- **Urban noise monitoring:** Helps identify and analyse common sound sources in cities.
- **Public safety support:** Detects critical events like sirens and alarms automatically.
- **Smart-city integration:** Useful for traffic analysis, incident detection, and noise management.
- **Data-driven planning:** Provides insights that aid quieter and more optimised city layouts.
- **Quality of life improvement:** Assists in reducing noise exposure near sensitive zones.

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APPENDIX – A

LIST OF FORMULAS USED (Urban Sound Classification Project)

1. Short-Time Fourier Transform (STFT)

$$\text{STFT}(x[n]) = \sum_{m=-\infty}^{\infty} x[m] w[n-m] e^{-j2\pi mn/N}$$

2. Power Spectrogram

$$P(f, t) = |\text{STFT}(f, t)|^2$$

3. Mel-Scale Conversion

$$\text{mel}(f) = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right)$$

4. MFCC Computation

(a) **Log Mel-Spectrogram**

$$S_{\text{mel}}(k) = \log \left(\sum_{f=0}^F P(f) \cdot M_k(f) \right)$$

(b) **Discrete Cosine Transform (MFCC)**

$$\text{MFCC}(n) = \sum_{k=1}^K S_{\text{mel}}(k) \cdot \cos \left(\frac{\pi n(k-0.5)}{K} \right)$$

5. Delta MFCC Coefficients

$$\Delta \text{MFCC}(t) = \frac{\sum_{n=1}^N n (\text{MFCC}(t+n) - \text{MFCC}(t-n))}{2 \sum_{n=1}^N n^2}$$

6. Zero Crossing Rate (ZCR)

$$\text{ZCR} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(x_t \cdot x_{t-1} < 0)$$

7. Batch Normalisation

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$y = \gamma \hat{x} + \beta$$

8. ReLU Activation Function

$$ReLU(x) = \max(0, x)$$

9. Softmax Activation

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

10. Categorical Cross-Entropy Loss

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

11. Adam Optimiser Update Rule

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \theta_t &= \theta_{t-1} - \alpha \frac{m_t}{\sqrt{v_t} + \epsilon} \end{aligned}$$

12. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

13. Precision

$$Precision = \frac{TP}{TP + FP}$$

14. Recall

$$Recall = \frac{TP}{TP + FN}$$

15. F1-Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

16. Confusion Matrix (General Form)

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

Confusion Matrix is an $N \times N$ matrix.

17. Min–Max Normalisation

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

18. Standardisation (Z-score Normalisation)

$$x_{\text{std}} = \frac{x - \mu}{\sigma}$$

19. Max Pooling Operation (CNN)

$$y = \max(x_1, x_2, \dots, x_n)$$

$$y = \max_{(i,j) \in W} x_{i,j}$$

20. Convolution Operation (CNN)

For 1D:

$$(f * g)(t) = \sum_{\tau} f(\tau) g(t - \tau)$$

For 2D:

$$Y(i, j) = \sum_m \sum_n X(i + m, j + n) K(m, n)$$



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